Severe community-acquired pneumonia

Severity

- CURB-65 ≥ 3 severe; PSI class IV–V
- IDSA/ATS major: septic shock on vasopressors or invasive ventilation; minor ≥ 3 (RR ≥ 30, P/F ≤ 250, multilobar, confusion, urea ≥ 20, WBC < 4, platelets < 100, T < 36, hypotension)

Microbiology

- Blood + sputum cultures, urinary pneumococcal and Legionella antigens, viral PCR, MRSA nasal swab
- Antibiotics: beta-lactam + macrolide (or respiratory quinolone); MRSA/Pseudomonas cover only if risk

Corticosteroids

- **CAPE COD 2023: hydrocortisone 200 mg/day for 8–14 days ↓ 28-day mortality** in severe CAP
- Avoid in influenza; benefit less clear in shock already on steroids

Complications

- ARDS, septic shock, empyema, abscess, AKI, MI/arrhythmia, AF

Key point: Severe CAP: severity → right antibiotics in 1 h → steroids → protect the lung.

HAP and VAP: diagnosis and treatment

Definitions

- HAP: pneumonia ≥ 48 h after admission; VAP: ≥ 48 h after intubation
- Diagnosis: new infiltrate + 2 of fever, purulent secretions, leucocytosis, worse oxygenation (CPIS > 6 supportive)

Sampling

- Non-invasive semi-quantitative ETA (≥ 10^5 CFU/mL) or BAL (≥ 10^4); mini-BAL; before antibiotics

Empirical antibiotics

- Anti-pseudomonal beta-lactam (pip-taz, cefepime, meropenem)
- Add MRSA cover (vancomycin/linezolid) if MRSA > 10–20% or prior MRSA
- Second anti-Gram-negative agent if septic shock, prior IV antibiotics in 90 days, high resistance

Duration

- **7 days** for most; de-escalate by culture; PCT helps stopping

Key point: Sample first, cover broadly, then narrow and stop at 7 days.

Notes: CAP ,TSS,vap, ARDS,AECOPD, P. effusions p.7

VAP prevention, SDD and oral care

Mechanical and positioning

- Head up 30–45°, cuff pressure 20–30 cmH2O, subglottic secretion drainage
- Avoid circuit changes unless soiled, closed suction, early tracheostomy not for VAP alone

Selective digestive decontamination

- Topical non-absorbable antibiotics (oropharynx + gut) ± short IV cefotaxime
- SuDDICU 2022: no significant mortality fall; lower bacteraemia; resistance concern in high-MDR settings

Other strategies

- Daily sedation holds and SBT, early extubation; avoid reintubation
- Chlorhexidine oral care: possible harm → toothbrushing preferred
- Stress-ulcer prophylaxis only if indicated (PPI ↑ pneumonia), probiotics not recommended

Key point: The best prevention is shortening ventilation.

ARDS: Berlin definition and lung-protective care

Berlin definition

- Within 1 week; bilateral opacities; not fully explained by cardiac failure/overload
- PaO2/FiO2 on PEEP ≥ 5: mild 200–300, moderate 100–200, severe ≤ 100 (2023 global: SpO2/FiO2, HFNC)

Phases

- Exudative (days 1–7: hyaline membranes, oedema) → proliferative → fibrotic

Lung-protective ventilation

- **Vt 6 mL/kg PBW, plateau ≤ 30, driving pressure ≤ 15** (ARMA)
- PEEP by FiO2 table or best compliance; conservative fluids (FACTT)
- Prone ≥ 16 h if P/F < 150 (PROSEVA); NMB not routine (ROSE)

Pharmacology

- Dexamethasone (DEXA-ARDS) considered; no benefit from statins, beta-agonists, surfactant

Key point: Keep driving pressure low: it predicts survival best.

Prone positioning and VV-ECMO

Prone physiology and evidence

- More homogeneous ventilation, recruits dorsal lung, better V/Q, less overdistension, RV unloading
- **PROSEVA: P/F < 150, ≥ 16 h/day → 28-day mortality 16% vs 33%**

Practice and complications

- Trained team (5–6), ETT secured, eyes/face/pressure areas, lines, feeding
- Complications: tube dislodgement, pressure sores, facial oedema, brachial plexus injury

VV-ECMO criteria (EOLIA)

- P/F < 50 for 3 h, < 80 for 6 h, or pH < 7.25 with PaCO2 ≥ 60 for 6 h despite optimal care
- EOLIA: 35% vs 46% (NS), Bayesian reanalysis favours ECMO; CESAR: referral to centre

Contraindications and outcome

- Relative: MV > 7 days, irreversible disease, contraindication to anticoagulation, severe brain injury
- RESP score predicts survival

Key point: Prone first; refer to an ECMO centre early when criteria approach.

AECOPD and NIV

Blood gas

- Acute-on-chronic respiratory acidosis: pH < 7.35, PaCO2 high, raised HCO3
- Cause: V/Q mismatch, ↑ dead space, muscle fatigue, dynamic hyperinflation

NIV

- Indication: pH < 7.35 and PaCO2 > 6.5 kPa despite medical therapy
- Contraindications: arrest, coma, vomiting, facial trauma, undrained pneumothorax, copious secretions
- Start IPAP 15 (up to 20–25), EPAP 4–5; recheck ABG 1–2 h; failure → intubate

Pharmacology

- Nebulised salbutamol + ipratropium, prednisolone 40 mg 5 days, antibiotics if purulent sputum
- Target SpO2 88–92%

If intubated

- Low RR, long expiratory time, avoid auto-PEEP, permissive hypercapnia, early NIV after extubation

Key point: NIV reduces intubation and mortality in hypercapnic AECOPD.

Notes: CAP ,TSS,vap, ARDS,AECOPD, P. effusions p.26

Life-threatening asthma and ventilation

Features

- Life-threatening: PEF < 33%, SpO2 < 92%, silent chest, cyanosis, exhaustion, arrhythmia, normal/raised PaCO2
- Near-fatal: raised PaCO2 and/or need for ventilation

Hypotension after intubation

- Dynamic hyperinflation (auto-PEEP) ± tension pneumothorax, hypovolaemia, sedatives
- **Disconnect from ventilator for 30–60 s**, chest compression, fluid, check for pneumothorax

Ventilator strategy

- Low RR (8–10), Vt 6–8 mL/kg, I:E 1:4–5, high inspiratory flow
- Plateau < 30, measure auto-PEEP, accept PaCO2 high if pH > 7.2

Drugs and rescue

- Salbutamol, ipratropium, IV hydrocortisone, IV magnesium 2 g
- Ketamine, aminophylline, adrenaline, volatile anaesthetics, ECCO2R/ECMO

Key point: In ventilated asthma, the key setting is a long expiratory time.

Notes: CAP ,TSS,vap, ARDS,AECOPD, P. effusions p.35

HFNO, ROX index and intubation timing

HFNO physiology

- Flow up to 60 L/min, FiO2 to 1.0; washes dead space, small PEEP, humidification, less work of breathing
- FLORALI: ↓ 90-day mortality vs NIV/standard O2 in de novo hypoxaemia

ROX index

- ROX = (SpO2/FiO2) / RR
- ≥ 4.88 at 2, 6, 12 h predicts success; < 3.85 high failure risk → intubate

Delayed intubation risks

- P-SILI from large swings, fatigue, emergency intubation, higher mortality (Kang 2015)

Safe intubation plan

- Preoxygenation with HFNO/NIV, RSI, video laryngoscope, fluids + vasopressor ready
- Experienced operator, checklist, apnoeic oxygenation

Key point: Reassess HFNO with ROX: don't let failing patients drift.

Notes: CAP ,TSS,vap, ARDS,AECOPD, P. effusions p.22

Pleural effusion and Light's criteria

Light's criteria (exudate if any)

- Fluid/serum protein > 0.5; fluid/serum LDH > 0.6; fluid LDH > 2/3 upper normal serum
- Parapneumonic: simple → complicated (pH < 7.2, glucose < 3.3 mmol/L, LDH > 1000) → empyema (pus)

Management

- Ultrasound-guided sampling; chest drain for complicated effusion/empyema
- Antibiotics with anaerobic cover; **intrapleural tPA + DNase (MIST-2)**; VATS if failing

Causes

- Transudate: heart failure, cirrhosis, nephrotic, hypoalbuminaemia, PD
- Exudate: pneumonia, malignancy, PE, TB, pancreatitis, autoimmune

Key point: pH < 7.2 in a parapneumonic effusion means drain it.

Notes: CAP ,TSS,vap, ARDS,AECOPD, P. effusions p.44

Massive haemoptysis

Definition and danger

- > 200–600 mL/24 h or any amount causing airway compromise or instability
- Death from asphyxia, not blood loss; bronchial arteries (systemic pressure) bleed in 90%

Immediate management

- Lie bleeding side down, high-flow O2, large-bore access, correct coagulation, TXA
- Intubate with large tube (≥ 8); single-lung intubation or bronchial blocker to isolate

Definitive work-up

- CT angiography to locate; bronchoscopy for localisation and local therapy
- **Bronchial artery embolisation** first-line; surgery for failure or localised lesion
- Causes: bronchiectasis, TB, cancer, aspergilloma, vasculitis, PE

Key point: Protect the good lung first: position and isolate.

Notes: CAP ,TSS,vap, ARDS,AECOPD, P. effusions p.19

Diffuse alveolar haemorrhage and pulmonary–renal syndrome

Diagnosis

- Haemoptysis (may be absent), falling Hb, diffuse infiltrates, hypoxaemia
- **BAL: progressively bloodier aliquots**, haemosiderin-laden macrophages
- Pulmonary–renal: ANCA vasculitis (GPA, MPA), anti-GBM (Goodpasture), SLE

Investigations

- ANCA (PR3, MPO), anti-GBM, ANA/dsDNA, complement, urine microscopy, renal biopsy
- Coagulation, echo, infection screen

Management

- High-dose methylprednisolone 1 g/day × 3 then cyclophosphamide or rituximab
- **Plasma exchange for anti-GBM**; considered in severe ANCA (PEXIVAS: no overall benefit)
- Lung-protective ventilation, RRT, correct coagulation, ECMO if refractory

Key point: Blood in the lungs + active urine sediment = pulmonary–renal syndrome.

Notes: chest continue p.1

Aspiration pneumonitis versus pneumonia

Differences

- Pneumonitis (Mendelson): chemical injury from gastric acid, witnessed, rapid onset in hours
- Pneumonia: oropharyngeal bacteria, dysphagia, slower onset, infiltrates in dependent segments

Antibiotics

- Not for pneumonitis in first 48 h unless SBO/gut obstruction or not resolving
- For pneumonia: amoxicillin-clavulanate or ceftriaxone; anaerobic cover not routine
- Stop meropenem/vancomycin if no risk factors (stewardship)

Supportive care and prevention

- Suction, O2, NIV/ventilation, no steroids
- Swallow assessment, head-up, oral care, avoid sedatives

Key point: Witnessed aspiration with rapid hypoxia is usually chemical: watch before treating.

Notes: chest continue p.9

Acute exacerbation of IPF

Definition

- Acute worsening < 1 month with new bilateral ground-glass on HRCT, not explained by heart failure/fluid
- Triggers: infection, aspiration, surgery, drugs, PE

Management

- Exclude infection, heart failure, PE; supportive O2/HFNO
- High-dose steroids commonly used (weak evidence); antifibrotics continued
- **Mechanical ventilation: hospital mortality > 80–90%**; only as bridge to transplant

Goals of care

- Early honest discussion: prognosis, ventilation limits, palliative care, DNACPR
- Patient's values and wishes, family involvement

Key point: AE-IPF has a poor prognosis: plan ceilings of care early.

Notes: chest continue p.12

Capnography in ICU

Physiology

- EtCO2 reflects production, delivery to lungs (CO) and ventilation; normal 35–45 mmHg; a–Et gap 2–5
- Waveform phases: I baseline, II rapid rise, III plateau, 0 inspiration

Applications

- Tube placement, CPR quality (> 20 mmHg) and ROSC, ventilation monitoring, transport
- Dead space and PE, sedation, cardiac output trend

Abnormal waveforms

- **Sudden EtCO2 fall + normal shape + hypotension = low cardiac output/PE**
- Zero: oesophageal intubation/disconnection; shark fin: bronchospasm; curare cleft: patient effort
- Rising baseline: rebreathing

Key point: Sudden EtCO2 drop with hypotension = circulatory collapse until proven otherwise.

Notes: chest continue p.19

Ventilator-induced lung injury and oxygen targets

VILI mechanisms

- Volutrauma (overdistension), barotrauma, atelectrauma (opening/closing), biotrauma (cytokines)
- Ergotrauma: mechanical power (RR, Vt, pressures, flow)

Ventilator changes

- Vt 6 mL/kg PBW, plateau ≤ 30, driving pressure ≤ 15, adequate PEEP, lower RR if possible

Oxygen toxicity

- Free radicals, absorption atelectasis, tracheobronchitis, hypercapnia
- **Target SpO2 92–96% / PaO2 8–10 kPa**; avoid hyperoxia (ICU-ROX, HOT-ICU, Mega-ROX)

Key point: Gentle ventilation and enough (not too much) oxygen protect the lung.

Notes: chest continue p.22

Patient–ventilator dyssynchrony

Classification

- Trigger: ineffective efforts, auto-triggering, double triggering, reverse triggering
- Flow: flow starvation; cycle: premature or delayed cycling

Causes and fixes

- Ineffective efforts: auto-PEEP, high support → reduce Vt/support, add PEEP
- Double triggering: Vt too small/short Ti → increase Ti or Vt
- Delayed cycling (COPD): raise cycle-off % (40–50%)

Sedation and new modes

- Treat pain/agitation; deep sedation masks, not fixes
- PAV+ and NAVA improve synchrony; NMB only if severe ARDS

Key point: Look at the waveforms before you increase sedation.

Notes: chest continue p.24

Tracheostomy: timing and emergencies

Indications and timing

- Prolonged MV, weaning failure, airway protection, secretions, upper airway obstruction
- Benefits: comfort, less sedation, oral hygiene, faster weaning
- **TracMan: early (≤ 4 days) vs late (≥ 10 days): no mortality difference**

Complications

- Early: bleeding, displacement, pneumothorax, surgical emphysema
- Late: stenosis, tracheo-innominate fistula, tracheo-oesophageal fistula, infection

Emergency (NTSP algorithm)

- Call help, O2 to face and stoma; remove speaking valve/inner tube; pass suction catheter
- Not patent: deflate cuff → remove tube → oxygenate via face/stoma → reintubate orally or stoma

Key point: A blocked trach: remove the inner tube first, then the trach if needed.

Notes: chest continue p.28

Pulmonary embolism with RV failure

Risk classes (ESC)

- High: shock/arrest; intermediate-high: RV dysfunction + raised troponin; intermediate-low; low (sPESI 0)
- Work-up: CTPA (or bedside echo if unstable), troponin, BNP, leg ultrasound

Reperfusion

- **High-risk: systemic thrombolysis** (alteplase 100 mg/2 h, or 0.6 mg/kg bolus in arrest)
- Intermediate-high: anticoagulate; rescue lysis if deterioration (PEITHO)
- Catheter-directed therapy or surgical embolectomy if lysis contraindicated/fails; PERT

RV support

- Avoid large fluids, noradrenaline for MAP, dobutamine, inhaled NO
- Avoid intubation if possible (RV collapse); VA-ECMO as rescue

Key point: In massive PE, don't fill the failing RV: give pressor and reperfuse.

Notes: chest continue p.13

Toxic shock syndrome

Diagnosis

- Fever, diffuse erythroderma, hypotension, ≥ 3 organ involvements, desquamation 1–2 weeks later
- Staphylococcal: tampons, wounds, blood cultures often negative, rash prominent
- Streptococcal: soft-tissue infection, pain, bacteraemia common, higher mortality

Management

- Remove source (tampon), drain/debride, fluids and vasopressors
- **Clindamycin (toxin suppression)** + anti-staph agent (flucloxacillin/vancomycin)
- Consider IVIG in severe/refractory cases

Organ dysfunction

- AKI, liver, DIC/thrombocytopenia, ARDS, encephalopathy, myositis (CK)

Key point: Superantigen shock: add clindamycin and remove the source.

Rhino-orbital-cerebral mucormycosis

Diagnosis and risks

- DKA/uncontrolled diabetes, steroids, neutropenia, iron overload, COVID-19
- Black eschar, facial swelling, proptosis, ophthalmoplegia, cranial nerve palsy

Confirmation

- **Tissue biopsy**: broad ribbon-like aseptate hyphae (right-angle branching), culture
- CT/MRI for orbital and cerebral spread

Management

- **Liposomal amphotericin B 5–10 mg/kg/day** + urgent surgical debridement
- Correct DKA/acidosis, stop steroids, isavuconazole/posaconazole step-down

Key point: Mucormycosis kills fast: biopsy, amphotericin and debridement without delay.

Notes: chest continue p.5